SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

FACULTY OF COMPUTING

IE3022

Applied Information Assurance

Assignment 01

IT22121592 - HERATH H.M.C.H.K

A Review of AI-Driven Anomaly Detection for Real-Time Monitoring of IoMT Networks

Author : Chameli Hansana , B.Sc. (Hons) in Information Technology, Sri Lanka Institute of Information Technology, Sri Lanka.
Email: chamelihansanaherath@gmaill.com

Abstract

The Internet of Medical Things (IoMT) is reshaping healthcare with real-time monitoring and connected smart devices. However, this connectivity brings new cybersecurity challenges. This review explores AI-driven techniques for real-time anomaly detection in IoMT networks, covering machine learning, deep learning, ensemble models, federated learning, and fuzzy logic. It also examines integrated security frameworks such as blockchain, edge/cloud computing, privacy-preserving protocols, and human-in-the-loop systems. Key challenges and future research needs—like lightweight models, data privacy, and adaptive detection—are discussed to guide the development of secure IoMT environments.

Index Terms— Internet of Medical Things (IoMT), Anomaly Detection, Intrusion Detection Systems, Machine Learning, Deep Learning, Ensemble Learning, Federated Learning, Fuzzy Logic, Blockchain, Edge Computing, Privacy Preservation, Cybersecurity

I. Introduction

The Internet of Medical Things (IoMT) revolutionizes modern healthcare through device connections between wearables sensors and medical equipment which enable real-time patient monitoring and data-driven decision-making. The advancement brings better diagnostics treatment accuracy and operational efficiency yet it creates essential security issues. The combination of restricted processing power and absent built-in security features in IoMT devices along with their transmission of sensitive health information makes them highly attractive to cybercriminals.

The complex and dynamic nature of these environments makes traditional security mechanisms based on rule or signature detection insufficient for protection. AI-driven anomaly detection systems have gained increasing attention because they detect abnormal behavior while adapting to new threats and providing immediate alerts through automated processes. Healthcare systems need to respect three essential constraints which include privacy concerns and both accuracy and real-time responsiveness requirements.

This review evaluates recent AI-based anomaly detection methods for IoMT networks while discussing their support from blockchain technology federated learning and edge/cloud architecture systems. The research targets researchers and practitioners to deliver an understandable summary of existing IoMT security methods and their current boundaries and upcoming developments.

II. Research Objectives

This literature review aims to critically analyze and synthesize current research about AI-driven anomaly detection for real-time IoMT network

monitoring. The review aims to achieve the following specific objectives.

- The research aims to group and explain AI-based anomaly detection techniques used in IoMT security which include machine learning and deep learning alongside ensemble models federated learning and fuzzy logic methods. The research summarizes the operational methods of each technique together with their reported success rates when detecting intrusions or anomalies within IoMT scenarios.

- The research investigates how anomaly detection techniques function when combined with complete security systems for IoMT networks. The review investigates how blockchain technology edge and cloud computing privacy-preserving protocols and human-in-the-loop systems support real-time anomaly detection in IoMT networks.

- The research identifies current limitations and unmet needs in IoMT anomaly detection systems. We identify key challenges which include data heterogeneity limited computational resources on IoMT devices and lack of labeled attack data high false alarm rates and scalability and deployment issues in healthcare environments.

- The research will present future research directions together with opportunities to enhance AI-driven IoMT anomaly detection. The research proposes several potential approaches which include lightweight models alongside explainable AI and enhanced federated learning schemes and multi-modal data fusion as solutions to address identified gaps for better IoMT network security.

The paper seeks to deliver researchers and practitioners a detailed understanding of current IoMT anomaly detection practices and future development directions for this essential cybersecurity field.

III. Review of Literature

A. Machine Learning-Based Anomaly Detection in IoMT

Some of the initial intrusion detection systems in the Internet of Medical Things (IoMT) applied traditional machine learning methods due to their capability of learning from data without being specifically programmed. Supervised learning approaches have been employed for the identification of network traffic or device behavior into normal or malicious classes. Common algorithms include support vector machines (SVMs), decision trees, random forests, K-nearest neighbors, and Naïve Bayes classifiers. The models work well with labeled training data and provide high accuracy under known attack patterns but perform poorly in generalizing to new attacks. Unsupervised learning and outlier detection methods are most effective in IoMT settings where obtaining labeled attack data is hard. Clustering-based anomaly detectors such as Isolation Forests and Local Outlier Factors (LOF) have been used in healthcare datasets to detect abnormal patterns. Tabassum et al. (2024) described an anomaly detection system for electronic health records using Isolation Forests and LOF which detected malicious insider activity with more than 99% accuracy and very low false positive rates . Such unsupervised models create a “normal” data profile (e.g., standard patient vital signs or network traffic metrics) and alert potential anomalies, which are suitable for detecting new or insider threats in smart healthcare environments.

IoMT can benefit from machine learning models because they are fast and simple, which can be beneficial for real-time operation on resource-constrained devices. They have

proven their effectiveness in numerous IoMT security researches. However, classical ML may not capture the complex temporal or spatial patterns in IoMT data as effectively as deep learning. A major problem with classical ML is the manual feature engineering (extracting the most important attributes from the raw data) which is quite a challenge when dealing with heterogeneous IoMT data (network logs, sensor readings, imaging data, etc.). To address these limitations, several recent works have shifted towards deep learning techniques to get improved accuracy and automated feature extraction.

B. Deep Learning Techniques for IoMT Intrusion Detection

The deep learning (DL) approaches use multi-layer neural networks which can automatically learn rich feature representations from raw data, which is beneficial for the complex and high-dimensional data in IoMT systems. Feed-forward dense networks, convolutional neural networks (CNNs), recurrent neural networks (RNNs) like Long Short-Term Memory (LSTM), and autoencoders have been applied by researchers to anomaly detection in IoMT. These models can capture nonlinear relationships and temporal dynamics inherent in IoMT data (e.g., sequences of network events or physiological signal patterns). Some studies have found that deep learning models are more effective than traditional methods in detecting IoMT attacks. Autoencoders have been employed to detect anomalies by learning to reconstruct normal behavior and identifying large reconstruction errors as potential intrusions. Manoharan and Thathan (2024) introduced a probabilistic deep autoencoder optimized with a group teaching algorithm for intrusion types classification in IoMT networks.

Their framework, tested on public IoT/IoMT datasets (BoT-IoT, ToN-IoT, etc.), got around 99% accuracy and F1-score for attacks vs. normal traffic, a substantial improvement over previous methods. Neural network classifiers and ensembles have also been shown to have high precision in identifying various cyber attacks on medical IoT devices. Sequential data can be modeled well by recurrent neural networks like LSTMs, which have been widely used for anomaly detection in network traffic and device telemetry. In IoMT, LSTM-based IDS can learn temporal patterns of sensor readings or communication sequences. Alalhareth and Hong (2023) introduced a self-tuning LSTM IDS enhanced with fuzzy logic for IoMT, which adaptively modified the training process to enhance performance. The resulting model had high accuracy and detection rates with almost no false positivesMDPI.COM

The study demonstrates how deep learning methods succeed in detecting IoMT attack signatures. Research has investigated hybrid CNN-LSTM models which utilize CNN to detect spatial patterns and LSTM to identify temporal patterns for monitoring system intrusion detection. These deep networks have the ability to detect unusual patient monitor readings which may indicate sensor tampering or network malware anomalies. In IoMT security, Explainable AI (XAI) techniques serve as a deep learning method for model decision interpretation. The research presented an explainable model named XSRU-IoMT which used simple recurrent units with skip connections to identify complex attacks and generate understandable explanations for each notification. The healthcare industry requires such functionality because clinicians need trust along with understanding the reasons behind alarm activations (for example, an anomaly detector alerting about a life-critical device). The adoption of deep learning in IoMT requires addressing both computational expenses and requirements for transparent systems.

Researchers address this limitation by running DL models at the edge or cloud layer of healthcare networks because many IoMT devices lack the capability to execute large neural networks locally. The paper establishes deep learning as a strong anomaly detection technique for IoMT because it delivers high accuracy and adaptability yet requires engineering to fulfill the domain-specific needs of real-time processing and interpretability.

C. Ensemble and Hybrid Models.

The combination of multiple algorithms through ensemble models leads to better detection accuracy while decreasing false positives in IoMT networks. Ensemble models leverage different classifiers including traditional ML and deep learning to achieve better generalization performance. The RKOA-AEID framework reaches high performance through its combination of feature selection and deep learning ensemble methods. Hybrid models combine AI with fuzzy logic and rule-based systems to improve their adaptability. The implementation of these methods requires detailed tuning to preserve efficiency on IoMT devices with limited resources although they deliver effective results**.**

D. Federated Learning for Distributed IoMT Anomaly Detection

Federated Learning (FL) represents an innovative method which tackles one of the main IoMT security problems by protecting patient privacy during collaborative learning. FL enables IoMT devices and edge nodes to develop local intrusion detection models through training their own data without sending raw medical information to central servers. The local models distribute their learned parameters for aggregation into a global model without exposing sensitive health information. The decentralized method keeps all health information at its origin while meeting HIPAA and GDPR data protection standards.

The healthcare sector has achieved promising results through frameworks like FedCure which implements personalization methods to modify the global model according to device-specific data distribution patterns. The technology finds great application in IoMT systems because these environments operate with diverse device characteristics and data types and attack exposure. The FedCure system merges distributed computing with customized distributed learning techniques to achieve both low latency and local pattern detection without compromising security.

The implementation of federated learning in IoMT systems encounters multiple difficulties despite its beneficial aspects. The system must handle non-IID data distributions between devices while managing communication overhead from model updates and addressing edge device computational constraints. The solution of these problems demands optimization approaches which include efficient aggregation algorithms together with compression techniques and lightweight model architectures.

Federated learning offers a strong method for developing privacy-focused intelligent anomaly detection systems in IoMT because the healthcare industry needs secure real-time monitoring solutions that scale.

E. Blockchain and Distributed Ledger Integration

The system requires trust and tamper-resistant data integrity to protect health data and security logs from malicious alteration in IoMT networks. The research community has investigated blockchain technology to establish a secure distributed ledger system for IoMT data which can enhance anomaly detection systems.

The tamper-proof nature of blockchain technology allows secure storage of IoMT data and anomaly logs. Healthcare systems use decentralized ledgers to establish real-time monitoring trust and data traceability which ensures data integrity. The decentralized ledger system enables real-time monitoring trust and data traceability. The combination of blockchain technology with edge computing systems creates frameworks which simultaneously protect privacy while minimizing latency. Hyperledger-based systems authenticate sensor data before it gets included in central health records. System

The widespread adoption of these systems faces two main ongoing obstacles which include system complexity and processing overhead..

F. Edge and Cloud Computing for Scalable Real-Time Detection.

The real-time anomaly detection in IoMT is enabled by both edge and cloud computing. The processing of data at the edge reduces latency and enables faster responses by handling information close to the source. However, the cloud infrastructure supports complex analytics and large-scale model training across healthcare networks. A hybrid edge–cloud approach enables scalable, low-latency detection of anomalies in high-volume IoMT data streams. However, the optimization of resource use and the maintenance of secure data transmission between system layers is still an ongoing challenge.

G. Privacy-Preserving Protocols and Data Protection.

The protection of medical data requires immediate attention in IoMT environments. The combination of encryption with differential privacy and secure federated learning enables both data protection against leakage and efficient anomaly detection. The analysis of data becomes possible through homomorphic encryption because it preserves the original form. The combination of edge processing with decentralized learning reduces the requirement for data transmission which leads to better privacy protection. Real-world healthcare settings require practical solutions which must properly manage privacy alongside model accuracy and system complexity.

H. Human-in-the-Loop Models

The combination of human operators with AI systems produces more precise and dependable anomaly detection results in IoMT networks. The human-in-the-loop (HiTL) models enable medical experts to confirm alerts while decreasing false positives and developing detection rules that reflect clinical expertise. The feedback loop enables ongoing learning and improves medical context alignment. The adoption of HiTL remains restricted but it demonstrates effective potential to enhance explainability and minimize unnecessary alarms in AI-based security systems.

Approach	Accuracy	Latency	Privacy	Real-world use
Traditional ML	Moderate	Low	Limited	Yes
Deep Learning	High	Medium	Low	Some
Ensemble Models	Very high	Medium	Low	Yes
Federated Learning	High	Low	High	Limited
Fuzzy Logic	Moderate	Medium	Medium	Some
Blockchain + AI	High	Medium	Very High	Emerging

IV. Future Research and Challenges

In spite of the progress in AI-driven anomaly detection for IoMT, several open challenges and research gaps remain. These challenges are important for developing the next generation of security solutions that are effective, efficient, and trustworthy in the healthcare domain. From the literature, the following are some of the challenges and future research directions that were identified.

Data Heterogeneity and Limited Attack Data: IoMT systems produce very diverse data including network traffic logs, sensor readings and electronic health records. Developing anomaly detection models that can integrate data from both the network and biomedical aspects remains a challenging task. Most of the current works either analyze network data alone or analyze each type of data in isolation. Future work should involve the investigation of multi-modal learning methods that integrate multiple data streams for comprehensive anomaly detection. Obtaining quality-labeled attack data in healthcare is challenging because attacks are rare and privacy is a concern. There is a need for IoMT-specific datasets and benchmarks that encompass various types of attacks including subtle and new attacks. It is possible to build good benchmarks if hospitals agree to share anonymized incident data with each other. Other approaches like data synthesis or simulation of medical IoT attacks may also be employed to enrich the training dataset. Standardizing evaluation methods across studies will enable fair comparison of different IDS approaches.

Lightweight and Real-Time Algorithms: This paper focuses on IoMT devices, which have limited resources, and therefore, require efficient anomaly detection algorithms. Although complex deep learning models can achieve high accuracy, they may not be practical to deploy on low-power devices or in situations where a rapid response is required. Future research should include model optimization as a technique to obtain smaller models without losing much accuracy through techniques like model compression, pruning, quantization, or knowledge distillation. An open area is the development of efficient algorithms such as one-class classifiers or incremental learners that can be executed on edge devices with low latency. Another related concept is that of edge intelligence which is the customisation of AI models for edge execution. Further, it is possible to investigate the possibility of on-device detection of some anomalies (like an unusual spike in a device's CPU usage which could indicate malware) with cloud-based analysis reserved for more complex patterns. Real-time detection is critical, and the future IDS should focus on quick data processing pipelines and may employ streaming analytics or online learning to detect and respond to threats in real-time.

Adversarial Resistance: AI models are themselves subject to adversarial attacks. Attackers could attempt to evade an IDS by creating inputs that deceive ML/DL models (for example, by modifying network traffic features in a subtle way) or by contaminating the training data in a federated learning scenario. The future should focus on making AI models resistant to adversarial manipulation.This includes adversarial training (training models on perturbed data), defensive distillation, and developing detection mechanisms for adversarial inputs. For IoMT, the high stakes mean an attacker who bypasses the IDS could do great harm, so making the IDS resilient is critical. The important research direction is to include adversarial resilience as a design principle, such as through monitoring of model confidence or through the use of anomaly detectors that are hard to spoof

Explainability and User Trust: As AI-driven IDS become more complex, for example, deep neural networks, their decisions may be hard to understand. In healthcare, it is specifically important to have explainable anomaly detection to allow security personnel and clinicians to believe and act on alerts. Future works should incorporate explainable AI (XAI) techniques, such as feature attribution methods (e.g., SHAP, LIME) or interpretable models, to give insights on why a specific device was flagged. It may point to a network packet or patient vital sign that caused the alert. Enhancing transparency will allow for better human-in-the-loop procedures and compliance with regulations (such as the need to justify decisions in clinical contexts). Research on generating human-friendly alerts (with risk levels or recommended actions) would make AI IDS more applicable to actual usage in the field.

The IoMT networks continue to develop with new technologies such as 5G/6G connectivity, Software-Defined Networking (SDN), and Network Function Virtualization (NFV). The use of each of them opens up new possibilities and problems for anomaly detection. 5G networks with their high speed ensure low latency but at the same time they increase the rate and the amount of data that needs to be monitored. Future research can consider the use of SDN based security, where the centralized control in SDN can be used for real time anomaly control, for example, reroute traffic in case of an anomaly. IDS functions can be deployed as virtualized services in NFV, which can scale on demand. It is important to study how anomaly detection algorithms can dynamically scale out or migrate in NFV/edge in order to protect IoMT devices. First studies have started to look at IDS in virtualized, cloud-edge configurations, but there is much more to explore in these paradigms. Furthermore, IDS and blockchain could be made to work together in a way that extends the use of blockchain from just logging to actually coordinating defense among devices. It is important to make sure that these integrations do not bring too much overhead.

Privacy and Ethical Considerations: As for the technical methods like encryption and federated learning which assist in preserving privacy, the future research should also address the healthcare regulations and patient consent for security monitoring. Patient related data inspected by IDS might have ethical concerns if not well managed. It is important to develop guidelines or frameworks that would ensure that anomaly detection is done in a privacy aware way, or at least, include privacy risk assessment for any new IDS deployment. Privacy preserving machine learning will remain an important area – techniques like secure multi-party computation for collaborative anomaly detection across institutions (without sharing raw data) are worth investigating. As mentioned in the literature, methods like

differential privacy can be implemented to give . The problem is how to find a balance between these privacy measures and the utility/accuracy of the security analytics.

Lifecycle Management and Updates: IoMT devices are built to last and are not often updated but the threat landscapes are constantly changing. Future research may investigate how to update anomaly detection models in the field. It could be done by online learning approaches that update the IDS model with each new data batch or detected event or by centralized periodic retraining with new threat intelligence. It also relates to human-in-the-loop: feedback loops for field experts to review and refine the detection rules on a regular basis. Another approach is transfer learning – if a model is trained on data from one hospital, how much of the learning can be transferred to another hospital with a little more training? Solving this can speed up the deployment of effective IDS across different IoMT environments.

In conclusion, AI driven anomaly detection for IoMT in the future will probably be more adaptive, collaborative, and transparent. It will continue to address the dual requirements of security and privacy. By focusing on lightweight design, robustness to adversaries, multi-modal analysis and human centric features (explainability and feedback), the next generation of IoMT IDS can achieve higher trust and effectiveness. Also, the integration of complementary technologies like blockchain, edge computing, etc. with AI will also define new solutions. Since IoMT devices are becoming even more prevalent in healthcare delivery, ensuring their security through intelligent anomaly detection will remain a critical and evolving frontier.

V. Conclusion

The Internet of Medical Things (IoMT) transforms healthcare delivery through its expanding connectivity which creates substantial cybersecurity threats. This review investigated AI-driven anomaly detection methods which include machine learning and deep learning together with federated learning and fuzzy logic to secure IoMT networks. The analysis included supporting technologies which improve detection and privacy and scalability through blockchain and edge/cloud computing and human-in-the-loop systems. The implemented techniques demonstrate strong potential but practical implementation faces obstacles related to data heterogeneity and resource constraints and model adaptability. Future research needs to develop lightweight and explainable collaborative solutions which meet clinical requirements. The security of IoMT needs immediate strengthening because it ensures patient safety and builds trust in healthcare systems that use connected technology.

VI. Acknowledgment

This paper acknowledges the academic and industry researchers whose valuable studies the author referenced in this paper. Their research has substantially improved the understanding of IoMT security. The Applied Information Assurance course faculty at SLIIT provided essential support which the author deeply appreciates. The author finished this work as an academic assignment while receiving valuable feedback and encouragement from instructors throughout the project.

VII. References

[1] Alruwaili, F. F., Asiri, M. M., Alrayes, F. S., Aljameel, S. S., Salama, A. S., & Hilal, A. M. (2023). Red Kite Optimization Algorithm With Average Ensemble Model for Intrusion Detection for Secure IoT. *IEEE Access*, *11*, 131749–131758. https://doi.org/10.1109/ACCESS.2023.3335124

[2] Lu, C., Lyu, J., Zhang, L., Gong, A., Fan, Y., Yan, J., & Li, X. (2020). Nuclear power plants with artificial intelligence in industry 4.0 era: Top-level design and current applications—a systemic review. In *IEEE Access* (Vol. 8, pp. 194315–194332). Institute of Electrical and Electronics Engineers Inc. https://doi.org/10.1109/ACCESS.2020.3032529

[3] Alruwaili, F. F., Asiri, M. M., Alrayes, F. S., Aljameel, S. S., Salama, A. S., & Hilal, A. M. (2023). Red Kite Optimization Algorithm With Average Ensemble Model for Intrusion Detection for Secure IoT. *IEEE Access*, *11*, 131749–131758. https://doi.org/10.1109/ACCESS.2023.3335124

[4] Syed, A., Ali, M., Ali, S., Ziaullah, K., Joo, M.-I., & Kim, H.-C. (n.d.). *Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000. IoMT and Blockchain Synergy: A Novel Approach to Health Data Validation and Storage*. https://doi.org/10.1109/ACCESS.2024.0429000

[5] Alruwaili, O., Wu, F., Mobarak, W., & Armghan, A. (2024). Enhancing Smart Factories Through Intelligent Measurement Devices Altering Smart Factories via IoT Infusion. *IEEE Access*, *12*, 45961–45975. https://doi.org/10.1109/ACCESS.2024.3382214

[6] Choyon, M. M. S., Rahman, M., Kabir, M. M., & Mridha, M. F. (2020). IoT based Health Monitoring Automated Predictive System to Confront COVID-19. *HONET 2020 - IEEE 17th International Conference on Smart Communities: Improving Quality of Life Using ICT, IoT and AI*, 189–193. https://doi.org/10.1109/HONET50430.2020.9322811

[7] Falowo, O. I., Botsyoe, L., Koshoedo, K., & Ozer, M. (2024). Enhancing Cybersecurity with Artificial Immune Systems and General Intelligence: A New Frontier in Threat Detection and Response. *IEEE Access*. https://doi.org/10.1109/ACCESS.2024.3454543

[8] Chen, P. Y., Cheng, Y. C., Zhong, Z. H., Zhang, F. Z., Pai, N. S., Li, C. M., & Lin, C. H. (2024). Information Security and Artificial Intelligence-Assisted Diagnosis in an Internet of Medical Thing System (IoMTS). *IEEE Access*, *12*, 9757–9775. https://doi.org/10.1109/ACCESS.2024.3351373

[9] Rakholia, R., Suárez-Cetrulo, A. L., Singh, M., & Carbajo, R. S. (2024). Advancing Manufacturing Through Artificial Intelligence: Current Landscape, Perspectives, Best Practices, Challenges and Future Direction. *IEEE Access*. https://doi.org/10.1109/ACCESS.2024.3458830

[10] Lins, R. G., & Givigi, S. N. (2021). Cooperative Robotics and Machine Learning for Smart Manufacturing: Platform Design and Trends within the Context of Industrial Internet of Things. *IEEE Access*, *9*, 95444–95455. https://doi.org/10.1109/ACCESS.2021.3094374

[11] Mahmoodi, A. B. Z., Sheikhi, S., Peltonen, E., & Kostakos, P. (2023). Autonomous Federated Learning for Distributed Intrusion Detection Systems in Public Networks. *IEEE Access*, *11*, 121325–121339.

https://doi.org/10.1109/ACCESS.2023.3327922

[12] Dutta, J., & Puthal, D. (2024). Advancing eHealth in Society 5.0: A Fuzzy Logic and Blockchain-Enhanced Framework for Integrating IoMT, Edge, and Cloud with AI. *IEEE Access*. https://doi.org/10.1109/ACCESS.2024.3520799

[13] Por, L. Y., Dai, Z., Leem, S. J., Chen, Y., Yang, J., Binbeshr, F., Phan, K. Y., & Ku, C. S. (2024). A Systematic Literature Review on the Methods and Challenges in Detecting Zero-Day Attacks: Insights from the Recent CrowdStrike Incident. In *IEEE Access*. Institute of Electrical and Electronics Engineers Inc. https://doi.org/10.1109/ACCESS.2024.3455410

[14] Fortoul-Diaz, J. A., Carrillo-Martinez, L. A., Centeno-Tellez, A., Cortes-Santacruz, F., Olmos-Pineda, I., & Flores-Quintero, R. R. (2023). A Smart Factory Architecture Based on Industry 4.0 Technologies: Open-Source Software Implementation. *IEEE Access*, *11*, 101727–101749. https://doi.org/10.1109/ACCESS.2023.3316116

[15] Sachin, D. N., Annappa, B., Hegde, S., Abhijit, C. S., & Ambesange, S. (2024). FedCure: A Heterogeneity-Aware Personalized Federated Learning Framework for Intelligent Healthcare Applications in IoMT Environments. *IEEE Access*, *12*, 15867–15883. https://doi.org/10.1109/ACCESS.2024.3357514

Author Profile

Chameli Hansana attends the Sri Lanka Institute of Information Technology (SLIIT) where she studies for her B.Sc. (Hons) in Information Technology with a concentration in Cyber Security. Her academic work centers on IoT security and intrusion detection systems and AI applications in healthcare cybersecurity. She finished work on secure IoT design and joined cyber defense competitions. Chameli plans to help create dependable medical technology security solutions through her work.

OneDrive Link:

IT22121592 IE3022